

DATA VISUALIZATION · COM-480 · EPFL 2025

CINESCOPE

What makes a movie successful?

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Milestone 2 · April 2025

ASTRA CINE - STARDUST DATA LOG

ASTRA CINE
STARDUST DATA PROJECT

Project Goal

CineScope explores a single overarching question: **what drives financial and critical success in cinema, and how have those relationships shifted across decades?** We treat financial and critical success as two parallel dimensions and examine where they align or diverge. Rather than presenting static conclusions, we build a coordinated interactive system that lets users test their own hypotheses across four linked views.

Following Milestone 1 feedback, we added a dedicated franchise view (View 04 Dynasties) examining whether sequel films reliably outperform their originals and under what genre and budget conditions.

Coordinated multiple-view design. A persistent genre filter and decade slider remain active across all four views simultaneously. Selecting a genre node in the Sankey diagram (View 02) filters the bubbles in the scatter (View 01); clicking a franchise point in the sequel scatter (View 04) highlights that film's position in the budget–revenue space. Every view is a lens on the same underlying question; users build their own narrative by linking observations across views.

Dataset. The Movies Dataset (Kaggle / TMDb): 45,697 films from 1874–2017 across 5 CSV files. After pre-processing, 5,417 films have valid budget and revenue figures. All visualizations draw from `movies_lite.json`, a browser-optimized export of the cleaned data.

Visualizations

View 01 — The Dossier Board

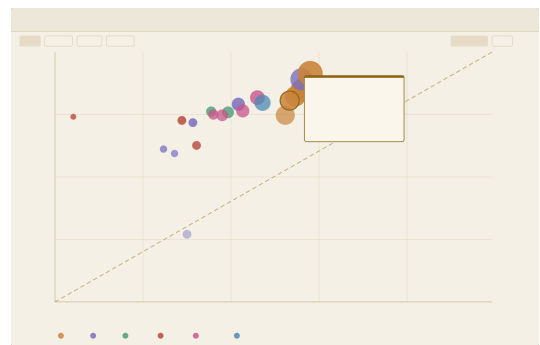
A **D3 scatter plot** where every film is a bubble. Color encodes genre; size encodes budget on a log–log scale. Hovering any bubble opens a film dossier card showing title, director, cast, budget, revenue, ROI badge, and rating. A y-axis toggle switches between box office revenue and ROI, offering two definitions of commercial success.

A dashed break-even diagonal and four quadrant annotations : *Blockbusters*, *Micro-budget Breakouts*, *Cult Losses*, *Big-budget Disappointments* that orient the reader immediately.

Interactions: genre pills, decade slider, lasso brush, y-axis toggle (Revenue / ROI), animated state transitions.

Tools: D3.js v7 scatter, D3 brush • **Lectures:** D3 Fundamentals (L4–5), Interaction (L6)

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View 02 — The Flow

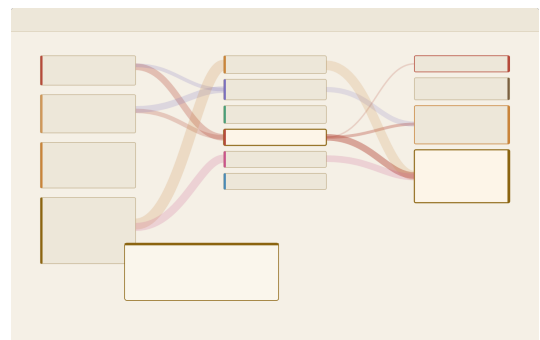
An **alluvial / Sankey diagram** mapping films from budget tier (*Micro* <\$5M, *Low*, *Mid*, *Blockbuster* >\$60M) through genre to financial outcome (*Flop*, *Break-Even*, *Hit*, *Megahit*). This single chart asks: does a given budget–genre pairing reliably flow toward profit?

Hovering a genre node highlights all paths through it, revealing patterns: Horror clusters in low-budget yet flows disproportionately into Hit outcomes; Animation needs blockbuster scale to reach Megahit. Clicking a genre cross-filters the bubbles in View 01.

Interactions: hover to highlight paths, click genre to cross-filter View 01, toggle revenue vs. ROI outcome tiers.

Tools: D3.js v7, d3-sankey plugin • **Lectures:** D3 Fundamentals (L4–5)

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View 03 — The Profitability Matrix

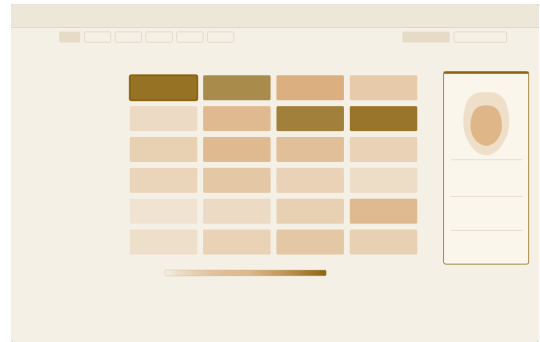
Core MVP · Functional

A **genre × budget-tier heatmap** where cell color encodes median ROI or probability of profit. Rows are genres (Action, Animation, Comedy, Drama, Horror, Thriller); columns are four budget tiers. A decade slider updates the matrix with a smooth D3 transition, showing how profitability has shifted from the 1970s to the 2010s.

Clicking any cell opens a **violin drill-down** showing the ROI distribution for that genre-budget pair against the full-dataset baseline. Key findings: Horror peaks at micro-budget (8.4×), Animation at blockbuster scale (8.1×), Comedy ROI declines sharply above mid-budget.

Interactions: decade slider, cell hover, click to violin drill-down, toggle Median ROI / % Profitable.

Tools: D3.js v7 heatmap, violin plot • **Lectures:** D3 (L4–5), Perception & Color (L7)



View 04 — Dynasties

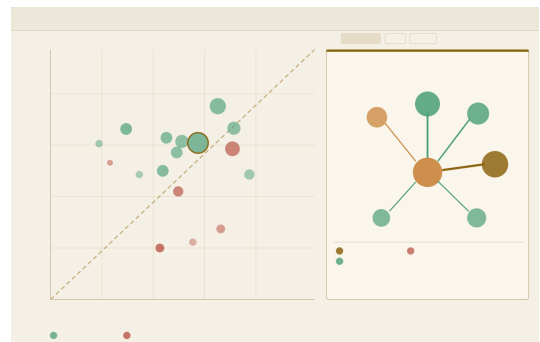
Core MVP · M1 Feedback · In Development

An **original vs. sequel revenue scatter**: x is the original's revenue, y is the first sequel's. The diagonal marks equal performance; points above improved, below declined. Color encodes percentage change (diverging: green / red); size encodes budget.

Clicking any franchise opens a **radial drill-down**: all installments branch from the original as nodes with size = revenue and branch thickness = budget, revealing whether a series peaks at part two or compounds across many films.

Interactions: filter by genre, decade, franchise size; hover card; click for radial drill-down; toggle revenue / ROI / rating.

Tools: D3.js v7 scatter, radial tree • **Lectures:** D3 Fundamentals (L4–5)



Tools and Lectures

View	Libraries & Technologies	Lectures
Dossier Board	D3.js v7 force simulation, D3 brush, CSS transitions	L4, L5, L6
The Flow	D3.js v7, d3-sankey plugin	L4, L5
Profitability Matrix	D3.js v7 heatmap, violin plot, D3 transitions	L4, L5, L7
Dynasties	D3.js v7 scatter, radial tree layout	L4, L5
Shared infrastructure	Vanilla JS, CSS custom properties, GitHub Pages	L1, L2, L3

Extra Features stretch goals that enhance without endangering the core

1. **Decade morphing animation** a play button on View 01 that animates scatter positions as the decade slider advances automatically, Gapminder-style.
2. **Persistent film search** a search bar fixed in the navigation that highlights a named film simultaneously across all four views.
3. **Parallel Coordinates Plot** additional view with axes for budget, runtime, TMDB rating, MovieLens rating, and ROI. Users brush multiple axes simultaneously to filter films. Kept as stretch.

Prototype

The prototype is live at [Cinescope](#) (GitHub Pages deployed on `main`).